

Solution 1

Question 1 (a)

Probability distribution P over binary variables $x_1, \dots, x_6 \in \{0, 1\}$ has the following form:

$$P(x_1, \dots, x_6) = \frac{1}{Z} \psi(x_1, x_2) \psi(x_2, x_3) \psi(x_3, x_4) \psi(x_4, x_5) \psi(x_5, x_6) \psi(x_6, x_1)$$

where for $a, b \in \{0, 1\}$,

$$\psi(a, b) = \begin{cases} 1 & \text{if } a = b \\ 0.5 & \text{otherwise} \end{cases}$$

Draw the minimal undirected graph G over which P factors. (This is the undirected graph with the fewest possible edges that contains cliques corresponding to each of the terms in P .)

Solution 1 (a)

Minimal Undirected Graph

The given probability distribution factors into pairwise potential functions $\psi(x_i, x_j)$. According to the Hammersley-Clifford theorem, a probability distribution factors over an undirected graph G if and only if it can be written as a product of potential functions over the maximal cliques of G . In this case, each potential function $\psi(x_i, x_{i+1})$ (with $x_7 = x_1$) corresponds to an edge between x_i and x_{i+1} in the graph. Since these potentials are pairwise, the graph is formed by directly connecting the variables involved in each potential.

The potential terms are:

- $\psi(x_1, x_2) \implies$ edge between x_1 and x_2
- $\psi(x_2, x_3) \implies$ edge between x_2 and x_3
- $\psi(x_3, x_4) \implies$ edge between x_3 and x_4
- $\psi(x_4, x_5) \implies$ edge between x_4 and x_5
- $\psi(x_5, x_6) \implies$ edge between x_5 and x_6
- $\psi(x_6, x_1) \implies$ edge between x_6 and x_1

This forms a cycle graph with 6 nodes.

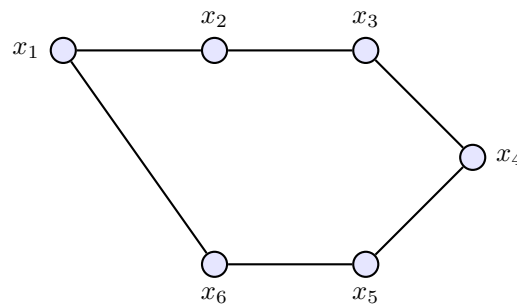


Figure 1: Minimal Undirected Graph for $P(x_1, \dots, x_6)$

$\therefore$ The minimal undirected graph is a 6-node cycle graph where each variable x_i is connected to x_{i-1} (or x_6 for $i = 1$) and x_{i+1} (or x_1 for $i = 6$).

Graduate Level Explanation

This problem concerns the graphical representation of a pairwise Markov Random Field (MRF), specifically an Ising-like model on a cycle graph. The factorization $P(\mathbf{x}) \propto \prod_{(i,j) \in E} \psi(x_i, x_j)$ directly implies that the graph $G = (V, E)$ must have edges corresponding to each pairwise potential function. For an MRF, the set of maximal cliques in the graph dictates the factorization of the joint probability distribution. In this case, each edge (x_i, x_{i+1}) forms a maximal clique of size two (a pairwise clique), thus forming a cycle graph. This structure directly encodes the local dependencies defined by the potential functions.

Simple Expedited Learning Explanation

Imagine each x_i is a person, and the $\psi(x_i, x_j)$ terms are "relationships" between pairs of people. If there's a relationship between x_1 and x_2 , you draw a line (an edge) connecting them. The problem lists six such relationships: x_1 with x_2 , x_2 with x_3 , and so on, until x_6 with x_1 . When you draw all these connections, starting from x_1 and going around to x_6 and back to x_1 , you end up with a closed loop, or a cycle graph. This graph is "minimal" because you only draw lines for the direct relationships given by the ψ terms, and no extra lines.

Solution 1

Question 1 (b)

Probability distribution P over binary variables $x_1, \dots, x_6 \in \{0, 1\}$ has the following form:

$$P(x_1, \dots, x_6) = \frac{1}{Z} \psi(x_1, x_2) \psi(x_2, x_3) \psi(x_3, x_4) \psi(x_4, x_5) \psi(x_5, x_6) \psi(x_6, x_1)$$

where for $a, b \in \{0, 1\}$,

$$\psi(a, b) = \begin{cases} 1 & \text{if } a = b \\ 0.5 & \text{otherwise} \end{cases}$$

For what configuration(s) $x \in \{0, 1\}^6$ is $P(x)$ maximized?

We want to find the configuration $x = (x_1, \dots, x_6)$ that maximizes $P(x)$.

$$\arg \max_x P(x) = \arg \max_x \left[\frac{1}{Z} \prod_{i=1}^5 \psi(x_i, x_{i+1}) \cdot \psi(x_6, x_1) \right]$$

Since the partition function Z is a positive constant, maximizing $P(x)$ is equivalent to maximizing the unnormalized product of potentials:

$$\arg \max_x P(x) = \arg \max_x [\psi(x_1, x_2) \psi(x_2, x_3) \psi(x_3, x_4) \psi(x_4, x_5) \psi(x_5, x_6) \psi(x_6, x_1)]$$

We are multiplying six potential terms together. From the definition of $\psi(a, b)$, the maximum possible value for any single potential is $\psi(a, a) = 1$. The other possible value is $\psi(a, b) = 0.5$ for $a \neq b$.

To maximize the product, we want to maximize the value of each term in the product. The largest possible product is achieved if we can make all six terms equal to 1. This requires:

- $\psi(x_1, x_2) = 1 \implies x_1 = x_2$
- $\psi(x_2, x_3) = 1 \implies x_2 = x_3$
- $\psi(x_3, x_4) = 1 \implies x_3 = x_4$
- $\psi(x_4, x_5) = 1 \implies x_4 = x_5$
- $\psi(x_5, x_6) = 1 \implies x_5 = x_6$
- $\psi(x_6, x_1) = 1 \implies x_6 = x_1$

Combining these conditions, we must have $x_1 = x_2 = x_3 = x_4 = x_5 = x_6$. Since the variables are binary, $x_i \in \{0, 1\}$, there are exactly two configurations that satisfy this condition:

1. $x = (0, 0, 0, 0, 0, 0)$
2. $x = (1, 1, 1, 1, 1, 1)$

For both of these configurations, the product of potentials is $1 \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot 1 = 1$. Any other configuration (e.g., $x = (0, 1, 0, 1, 0, 1)$) must have at least one pair of disagreeing neighbors. For every such pair, the potential function evaluates to 0.5. This will result in a total product strictly less than 1 (e.g., 0.5^6 for the alternating case). Therefore, the two configurations where all variables are aligned are the unique maximizers of $P(x)$.

$\therefore$ The two configurations that maximize $P(x)$ are $x = (0, 0, 0, 0, 0, 0)$ and $x = (1, 1, 1, 1, 1, 1)$.

Graduate Level Explanation

This problem is a Maximum A Posteriori (MAP) inference task on a pairwise Markov Random Field. The potential function $\psi(a, b)$, which assigns a higher value to agreement ($a = b$), defines this as a **ferromagnetic Ising model**. Finding the configuration x that maximizes $P(x)$ is equivalent to finding the **ground state** (minimum energy configuration) of the system. The probability $P(x) \propto \exp(-E(x))$ where $E(x) = -\sum_{(i,j)} \log \psi(x_i, x_j)$ is the energy. To maximize $P(x)$, we must minimize $E(x)$. Since $-\log(1) = 0$ and $-\log(0.5) > 0$, the energy is minimized (at $E = 0$) when all neighboring "spins" are aligned. This model exhibits two degenerate ground states: all spins "down" (all 0s) and all spins "up" (all 1s).

Simple Expedited Learning Explanation

Think of this as six friends sitting in a circle, and each must choose a "0" or a "1" hat. The "happiness" of the group is determined by multiplying "agreement scores" between all neighbors. If two neighbors have the same hat, their score ψ is 1. If they have different hats, their score is only 0.5. To make the **total** group happiness (the final $P(x)$) as high as possible, you need every single pair of neighbors to have the highest possible score. The highest score is 1, which happens when they match. The only way for everyone to match their neighbor in a circle is if everyone makes the same choice: either everyone wears a "0" hat, or everyone wears a "1" hat.

Solution 1

Question 1 (c)

Probability distribution P over binary variables $x_1, \dots, x_6 \in \{0, 1\}$ has the following form:

$$P(x_1, \dots, x_6) = \frac{1}{Z} \psi(x_1, x_2) \psi(x_2, x_3) \psi(x_3, x_4) \psi(x_4, x_5) \psi(x_5, x_6) \psi(x_6, x_1)$$

where for $a, b \in \{0, 1\}$,

$$\psi(a, b) = \begin{cases} 1 & \text{if } a = b \\ 0.5 & \text{otherwise} \end{cases}$$

For what configuration(s) $x \in \{0, 1\}^6$ is $P(x)$ minimized?

Solution 1 (c)

We want to find the configuration $x = (x_1, \dots, x_6)$ that minimizes $P(x)$.

$$\arg \min_x P(x) = \arg \min_x \left[\frac{1}{Z} \prod_{i=1}^5 \psi(x_i, x_{i+1}) \cdot \psi(x_6, x_1) \right]$$

Since Z is a positive constant, minimizing $P(x)$ is equivalent to minimizing the unnormalized product of potentials:

$$\arg \min_x P(x) = \arg \min_x [\psi(x_1, x_2) \psi(x_2, x_3) \psi(x_3, x_4) \psi(x_4, x_5) \psi(x_5, x_6) \psi(x_6, x_1)]$$

The potential function $\psi(a, b)$ has a minimum value of 0.5, which occurs when $a \neq b$. The value is 1 when $a = b$.

To minimize the product of these six terms, we want to make each term in the product as small as possible. The smallest possible product is achieved if we can make all six terms equal to 0.5. This requires that all adjacent nodes disagree:

- $\psi(x_1, x_2) = 0.5 \implies x_1 \neq x_2$
- $\psi(x_2, x_3) = 0.5 \implies x_2 \neq x_3$
- $\psi(x_3, x_4) = 0.5 \implies x_3 \neq x_4$
- $\psi(x_4, x_5) = 0.5 \implies x_4 \neq x_5$
- $\psi(x_5, x_6) = 0.5 \implies x_5 \neq x_6$
- $\psi(x_6, x_1) = 0.5 \implies x_6 \neq x_1$

This set of conditions requires a perfectly alternating assignment of values. Let's see if this is possible.

1. Case 1: Start with $x_1 = 0$

- $x_1 = 0 \implies x_2 = 1$
- $x_2 = 1 \implies x_3 = 0$
- $x_3 = 0 \implies x_4 = 1$
- $x_4 = 1 \implies x_5 = 0$
- $x_5 = 0 \implies x_6 = 1$
- **Check final link:** We need $x_6 \neq x_1$. Here, $x_6 = 1$ and $x_1 = 0$. This condition is satisfied.
- This gives the configuration: $x = (0, 1, 0, 1, 0, 1)$.

2. Case 2: Start with $x_1 = 1$

- $x_1 = 1 \implies x_2 = 0$
- $x_2 = 0 \implies x_3 = 1$
- $x_3 = 1 \implies x_4 = 0$
- $x_4 = 0 \implies x_5 = 1$
- $x_5 = 1 \implies x_6 = 0$
- **Check final link:** We need $x_6 \neq x_1$. Here, $x_6 = 0$ and $x_1 = 1$. This condition is satisfied.
- This gives the configuration: $x = (1, 0, 1, 0, 1, 0)$.

These two alternating configurations both result in a product of potentials equal to $0.5^6 = 0.015625$. Any other configuration must have at least one pair of agreeing neighbors, e.g., $\psi(x_i, x_{i+1}) = 1$. This would make the product $1 \cdot 0.5^5 = 0.03125$ (or larger), which is greater than 0.5^6 . Therefore, the two alternating configurations are the unique minimizers.

$\therefore$ The two configurations that minimize $P(x)$ are the alternating patterns: $x = (0, 1, 0, 1, 0, 1)$ and $x = (1, 0, 1, 0, 1, 0)$.

Graduate Level Explanation

This problem asks for the maximum energy (minimum probability) states of a ferromagnetic Ising model. The potential $\psi(a, b)$ penalizes disagreement, so the highest energy states are those with the most disagreements. The underlying graph is a 6-cycle, which is a **bipartite graph** (a graph whose vertices can be divided into two disjoint sets such that every edge connects a vertex in one set to one in the other). For a bipartite graph, a 2-coloring is possible, which corresponds exactly to an assignment where no two adjacent nodes have the same value. The two configurations found, $(0, 1, 0, 1, 0, 1)$ and $(1, 0, 1, 0, 1, 0)$, represent the two valid 2-colorings of this graph. These states maximize the system's energy (minimize probability) by maximizing the number of "unhappy" (disagreeing) links. This is only possible without **frustration** because the cycle length is even.

Simple Expedited Learning Explanation

Following the "six friends in a circle" analogy, we now want to find the configuration that makes the group **unhappiest**. The group's "happiness score" is 1 if neighbors match and only 0.5 if they differ. To get the lowest possible total score (the minimum $P(x)$), we need every single pair of neighbors to get the lowest score of 0.5. This means everyone must have a different hat from their neighbors. If person 1 picks "0", person 2 must pick "1", person 3 must pick "0", and so on. This creates an alternating pattern $(0, 1, 0, 1, 0, 1)$. We check the last link: person 6 (with "1") is different from person 1 (with "0"), so it works! The only other possibility is starting with "1", which gives the other alternating pattern $(1, 0, 1, 0, 1, 0)$.

Solution 2

Question 2 (a)

In a tournament, each player plays three games. Define three random variables:

- Random variable A is 1 if Tom wins his first game, and is 0 otherwise.
- Random variable B is 1 if Tom wins his first two games, and is 0 otherwise.
- Random variable C is 1 if Tom wins all three of his games, and is 0 otherwise.

Is C independent of A (i.e., is it true that for any values a, c that A, C might take on, we have $\Pr(A = a, C = c) = \Pr(A = a) \Pr(C = c)$)? Give a brief (one-sentence) reason for your answer.

Hint: The answer is no; show this by exhibiting specific choices of a, c for which $\Pr(A = a, C = c) \neq \Pr(A = a) \Pr(C = c)$.

Solution 2 (a)

We follow the hint and test the case where $a = 0$ and $c = 1$.

- $A = 0$ is the event "Tom does not win his first game."
- $C = 1$ is the event "Tom wins all three of his games."

By definition, the event $C = 1$ implies that Tom wins Game 1, Game 2, and Game 3. This means that $C = 1$ and $A = 0$ are mutually exclusive events; it is impossible for Tom to lose his first game *and* win all three games.

Therefore, the joint probability is $\Pr(A = 0, C = 1) = 0$.

Now we consider the product of the marginal probabilities, $\Pr(A = 0) \Pr(C = 1)$. Let's assume the probability of winning any game i is p_i , where $0 < p_i < 1$. This means the games are non-deterministic (it's possible to both win and lose).

- $\Pr(A = 0) = \Pr(\text{Lose Game 1}) = 1 - p_1$. Since $p_1 < 1$, we have $\Pr(A = 0) > 0$.
- $\Pr(C = 1) = \Pr(\text{Win G1, G2, G3})$. Assuming game independence, $\Pr(C = 1) = p_1 p_2 p_3$. Since $p_i > 0$, we have $\Pr(C = 1) > 0$.

The product of the marginals is $\Pr(A = 0) \Pr(C = 1) = (1 - p_1)(p_1 p_2 p_3)$, which is strictly greater than 0.

We have found a case where $\Pr(A = 0, C = 1) = 0$ but $\Pr(A = 0) \Pr(C = 1) > 0$. Since $0 \neq (1 - p_1)(p_1 p_2 p_3)$, the independence condition $\Pr(A = a, C = c) = \Pr(A = a) \Pr(C = c)$ is violated.

$\therefore$ No, C is not independent of A because if Tom loses his first game ($A = 0$), he *cannot* win all three games ($C = 1$), so $\Pr(A = 0, C = 1) = 0$, which is not equal to the product $\Pr(A = 0) \Pr(C = 1)$ (as both marginal probabilities are non-zero, assuming non-trivial games).

Graduate Level Explanation

The variables A and C are not marginally independent because a deterministic, causal relationship exists between them. Specifically, C is a logical subset of A ; the event $C = 1$ (winning all three) *implies* the event $A = 1$ (winning the first). Therefore, $\Pr(C = 1 \mid A = 0) = 0$. For independence to hold, this conditional probability must equal the marginal probability $\Pr(C = 1)$, but $\Pr(C = 1) > 0$ (assuming a non-zero chance of winning). Because $\Pr(C = 1 \mid A = 0) \neq \Pr(C = 1)$, they are dependent; observing $A = 0$ provides definitive information about C (namely, that $C = 0$).

Simple Expedited Learning Explanation

No. Think of it this way: "Winning all three games" ($C = 1$) *includes* "Winning the first game" ($A = 1$). They are not separate events. If I tell you "Tom lost the first game" ($A = 0$), you know *for sure* that "Tom won all three games" ($C = 1$) is false. Since knowing the outcome of A gives you 100% certainty about C in this case, they are clearly dependent.

Solution 2

Question 2 (b)

In a tournament, each player plays three games. Define three random variables:

- Random variable A is 1 if Tom wins his first game, and is 0 otherwise.
- Random variable B is 1 if Tom wins his first two games, and is 0 otherwise.
- Random variable C is 1 if Tom wins all three of his games, and is 0 otherwise.

Is C conditionally independent of A given B (i.e., is it true that $\Pr(C = c | A = a, B = b) = \Pr(C = c | B = b)$)? Give a brief reason.

Solution 2 (b)

Yes, C is conditionally independent of A given B . We can prove this by checking all possible values of the conditioning variable B .

Let $p_3 = \Pr(\text{Win Game 3} \mid \text{Win Game 1, Win Game 2})$. We assume $0 < p_3 < 1$.

1. Case 1: We condition on $B = 1$ (Tom wins the first two games).

- If $B = 1$, then A *must* be 1 (winning the first two games implies winning the first game). The case $A = 0, B = 1$ has zero probability, so we only need to check $A = 1, B = 1$.
- $\Pr(C = 1 \mid B = 1)$: Given Tom won the first two games, the probability of winning all three is just the probability of winning Game 3. So, $\Pr(C = 1 \mid B = 1) = p_3$.
- $\Pr(C = 1 \mid A = 1, B = 1)$: Given Tom won the first game ($A = 1$) AND won the first two games ($B = 1$), the $A = 1$ information is redundant. The probability of winning all three is still just the probability of winning Game 3. So, $\Pr(C = 1 \mid A = 1, B = 1) = p_3$.
- In this case, $\Pr(C \mid A, B = 1) = \Pr(C \mid B = 1)$.

2. Case 2: We condition on $B = 0$ (Tom does not win the first two games).

- If Tom did not win the first two games, it is impossible for him to win all three games.
- $\Pr(C = 1 \mid B = 0) = 0$.
- Now we check $\Pr(C = 1 \mid A, B = 0)$ for both values of A :
- $\Pr(C = 1 \mid A = 0, B = 0)$: Given Tom lost the first game ($A = 0$), it is impossible to win all three. This probability is 0.
- $\Pr(C = 1 \mid A = 1, B = 0)$: Given Tom won the first game ($A = 1$) but failed to win the first two ($B = 0$) (i.e., he lost Game 2), it is impossible to win all three. This probability is 0.
- In all sub-cases where $B = 0$, $\Pr(C = 1 \mid A, B = 0) = 0$, which equals $\Pr(C = 1 \mid B = 0)$.

Since $\Pr(C \mid A, B) = \Pr(C \mid B)$ holds for all possible values of A, B, C , the conditional independence statement is true.

$\therefore$ Yes, C is conditionally independent of A given B , because once we know the outcome of the first two games (B), knowing the outcome of *only* the first game (A) provides no new information for predicting the outcome of all three games (C).

Graduate Level Explanation

Yes. The probabilistic relationships $A \leftarrow B \leftarrow C$ form a **Markov chain** (where $C = 1 \implies B = 1 \implies A = 1$). A fundamental property of Markov chains is that, given the "present" state (B), the "future" (C) is conditionally independent of the "past" (A). The variable B acts as an information bottleneck; it "screens off" any influence A has on C . Any path from A to C in the dependency graph must pass through B , so conditioning on B blocks this path, satisfying the definition of d-separation.

Simple Expedited Learning Explanation

Yes. Think of B as a "gate". To predict C (winning all 3 games), you only need to know two things:

1. Did Tom pass the gate B (win the first two games)?
2. Did Tom win Game 3?

If I tell you $B = 0$ (he didn't pass the gate), you know C is 0 (he can't win all 3). It doesn't matter **why** he failed B (whether he lost G1 or G2). If I tell you $B = 1$ (he passed the gate), all you care about now is Game 3. Does it help to **also** tell you $A = 1$ (he won Game 1)? No, you already knew that, because $B = 1$ **means** he won Game 1 and Game 2. Since B tells you everything A could, and more, B makes A redundant.

Solution 2

Question 2 (c)

In a tournament, each player plays three games. Define three random variables:

- Random variable A is 1 if Tom wins his first game, and is 0 otherwise.
- Random variable B is 1 if Tom wins his first two games, and is 0 otherwise.
- Random variable C is 1 if Tom wins all three of his games, and is 0 otherwise.

Draw an undirected graphical model for (A, B, C) .

Solution 2 (c)

Solution 2 (c)

Undirected Graphical Model

An undirected graphical model (UGM) represents conditional independencies. An edge is absent between two nodes X and Y if they are conditionally independent given all other variables in the graph.

From our answer in 2(b), we proved the key conditional independence relationship: $A \perp C \mid B$. This means that in our undirected graph, the node B must separate nodes A and C . The only way to achieve this in a three-node graph is to *not* have a direct edge between A and C .

We must also check if other edges are required:

- **Edge (A, B):** Are A and B conditionally independent given C ? No. For example, if $C = 0$, knowing $A = 0$ makes $B = 0$ certain, while knowing $A = 1$ does not (B could be 0 or 1, though $B = 1$ is impossible if $C = 0$). Since $A \not\perp B \mid C$, we need an edge between A and B .
- **Edge (B, C):** Are B and C conditionally independent given A ? No. If $A = 1$, knowing $B = 0$ (lost G2) makes $C = 0$ certain, while knowing $B = 1$ (won G1 & G2) makes C uncertain (depends on G3). Since $B \not\perp C \mid A$, we need an edge between B and C .

The graph must therefore have edges (A, B) and (B, C) , but no edge (A, C) . This forms a simple chain.

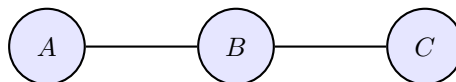


Figure 2: Undirected Graphical Model for (A, B, C)

$\therefore$ The undirected graphical model is the chain $A - B - C$, which correctly encodes the relationship that A and C are conditionally independent given B .

Graduate Level Explanation

The minimal directed acyclic graph (DAG) that captures the dependency structure is the Markov chain $A \rightarrow B \rightarrow C$. This is because the event B (winning G1 & G2) depends on A (winning G1), and C (winning G1, G2, & G3) depends on B . This DAG encodes the conditional independence $A \perp C \mid B$ via d-separation. To find the minimal undirected graph (the "moral graph") that preserves this independence, we apply the moralization procedure: 1) "Marry" (connect) all pairs of non-adjacent parents, and 2) Drop all arrowheads. Since there are no nodes with multiple parents (no v-structures), no edges are added. Dropping the arrows from $A \rightarrow B \rightarrow C$ directly yields the UGM $A - B - C$.

Simple Expedited Learning Explanation

Think of the graph as showing who "talks to" whom.

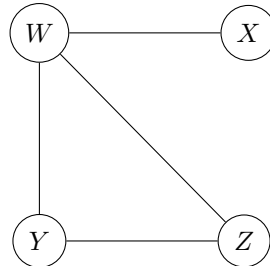
- A (Win G1) talks directly to B (Win G1 & G2), because B 's outcome is built from A 's.
- C (Win G1, G2, & G3) talks directly to B (Win G1 & G2), because C 's outcome is built from B 's.
- A and C do *not* talk directly. C gets all the information it needs about A *through* B .

Therefore, we draw a line from A to B , and from B to C . The resulting picture is just a simple chain: $A - B - C$. The missing line between A and C is the most important part: it means "if you know B , A and C have nothing left to say to each other."

Solution 3

Question 3 (a)

Four random variables W, X, Y, Z factor over the graph shown below.



What is the functional form of the joint distribution $P(W, X, Y, Z)$?

Solution 3 (a)

Solution 3 (a)

According to the **Hammersley-Clifford theorem**, any positive probability distribution P that factors over an undirected graph G can be written as a product of potential functions (ψ_C) over the **maximal cliques** of the graph, normalized by a partition function Z .

1. **Define a Clique:** A clique is a subset of nodes in the graph where every node in the subset is connected to every other node in the subset.
2. **Define a Maximal Clique:** A clique is **maximal** if it is not a subset of any larger clique.
3. **Identify Cliques in G:**

- *Size 2 (edges):* $\{W, X\}, \{W, Y\}, \{W, Z\}, \{Y, Z\}$
- *Size 3 (triangles):* We check subsets of size 3.
 - $\{W, Y, Z\}$: The edges are (W, Y) , (W, Z) , and (Y, Z) . All three edges exist. This is a clique.
 - $\{W, X, Y\}$: The edges are (W, X) , (W, Y) , and (X, Y) . The edge (X, Y) is missing. This is not a clique.
 - $\{W, X, Z\}$: The edges are (W, X) , (W, Z) , and (X, Z) . The edge (X, Z) is missing. This is not a clique.
- *Size 4:* $\{W, X, Y, Z\}$ is not a clique (e.g., (X, Y) is missing).

4. **Identify Maximal Cliques $\mathcal{C}$:**

- Is $\{W, Y, Z\}$ maximal? Yes. It is a clique, and we cannot add the only remaining node X (as X is not connected to Y or Z).
- Is $\{W, X\}$ maximal? Yes. It is a clique, and we cannot add Y (no $X - Y$ edge) or Z (no $X - Z$ edge).
- The other cliques, $\{W, Y\}$, $\{W, Z\}$, and $\{Y, Z\}$, are **not** maximal because they are all subsets of the larger clique $\{W, Y, Z\}$.

5. **Construct Functional Form:** The set of maximal cliques is $\mathcal{C} = \{\{W, X\}, \{W, Y, Z\}\}$. The joint distribution is the product of potentials over these two cliques.

$\therefore$ The functional form is $P(W, X, Y, Z) = \frac{1}{Z} \psi_1(W, X) \psi_2(W, Y, Z)$, where Z is the partition function, defined as $Z = \sum_{w, x, y, z} \psi_1(w, x) \psi_2(w, y, z)$.

Graduate Level Explanation

The Hammersley-Clifford theorem establishes the equivalence between the conditional independence properties of an undirected graphical model (a Markov Random Field) and its factorization into a product of potentials over its maximal cliques. This form of distribution is also known as a **Gibbs distribution**. The graph's topology, $G = (V, E)$, defines the set of maximal cliques $\mathcal{C}$. Here, $\mathcal{C} = \{\{W, X\}, \{W, Y, Z\}\}$. This factorization implies all the conditional independencies in the graph, such as $X \perp \{Y, Z\} \mid W$, which can be verified by graph separation.

Simple Expedited Learning Explanation

Think of the graph as a network of "friend groups" where everyone in a group is friends with everyone else. These groups are "cliques." We are looking for the **biggest** friend groups that can't add any new members (these are "maximal cliques").

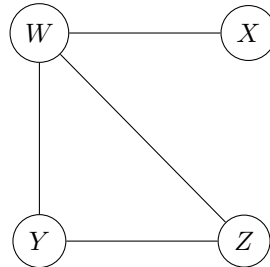
- W, Y, Z are all friends with each other, forming a trio. They can't add X because X isn't friends with Y or Z . So, $\{W, Y, Z\}$ is one maximal group.
- W, X are friends. They can't add Y (not friends with X) or Z (not friends with X). So, $\{W, X\}$ is another maximal group.

The total probability of the system is just the "score" of the $\{W, X\}$ group multiplied by the "score" of the $\{W, Y, Z\}$ group, all divided by a number Z to make sure the probabilities add up to 1.

Solution 3

Question 3 (b)

Four random variables W, X, Y, Z factor over the graph shown below.



Write down a conditional independence property of this distribution.

Solution 3 (b)

In an undirected graphical model, conditional independence is determined by **graph separation**. A set of nodes A is conditionally independent of a set of nodes B given a set of nodes C (written $A \perp B \mid C$) if C separates A and B . This means that every path from any node in A to any node in B must pass through at least one node in C .

Let's test the independence between X and the set $\{Y, Z\}$.

- Let $A = \{X\}$.
- Let $B = \{Y, Z\}$.
- Let's test if $C = \{W\}$ is a separating set.

We must check all paths from A to B :

1. **Paths from X to Y :** The two paths are $X - W - Y$ and $X - W - Z - Y$. Both of these paths pass through W .
2. **Paths from X to Z :** The two paths are $X - W - Z$ and $X - W - Y - Z$. Both of these paths also pass through W .

Since every possible path from X to any node in $\{Y, Z\}$ is blocked by (i.e., must pass through) the node W , the set $C = \{W\}$ separates $A = \{X\}$ from $B = \{Y, Z\}$.

Therefore, the distribution must satisfy the conditional independence property $X \perp \{Y, Z\} \mid W$.

$\therefore$ A conditional independence property of this distribution is $X \perp \{Y, Z\} \mid W$ (or $X \perp Y \mid W$, or $X \perp Z \mid W$, which are also true).

Graduate Level Explanation

The graph G implies a set of conditional independencies defined by the **Global Markov Property**: for any disjoint sets A, B, C , if C separates A from B , then $A \perp B \mid C$. The node W acts as an **articulation point** (or separating set) between node X and the subgraph induced by $\{Y, Z\}$. The factorization found in 3(a), $P \propto \psi_1(W, X)\psi_2(W, Y, Z)$, also implies this. When we condition on W , the joint distribution $P(X, Y, Z \mid W)$ becomes proportional to $\psi_1(W, X)\psi_2(W, Y, Z)$. Since X only appears in the first potential and $\{Y, Z\}$ only in the second, the distribution factors as $P(X \mid W)P(Y, Z \mid W)$, which is the definition of $X \perp \{Y, Z\} \mid W$.

Simple Expedited Learning Explanation

Think of the graph lines as information channels. X can only send or receive information to Y or Z by going *through* W . The node W is the "gatekeeper" for X . If we "condition on W ", it means we are standing at the gate W and we already know its value. Once we know W 's value, X has no *new* information to tell Y or Z , and Y and Z have no *new* information to tell X . By observing the gatekeeper W , we "block" the channel of influence, making X and the pair $\{Y, Z\}$ independent of each other.

Solution 4

Question 4 (a)

The joint distribution P over (x_1, x_2, x_3) is given by the following table.

x_1	x_2	x_3	Pr
0	0	0	1/3
0	0	1	1/3
1	0	0	1/6
1	1	1	1/6

One of these variables is a deterministic function of the other two. Identify this relationship.

Solution 4 (a)

A variable Y is a deterministic function of X and Z if, for every pair of values (x, z) that has a non-zero probability of occurring, there is only one possible value for y . We inspect the four configurations in the table that have non-zero probability.

1. **Is $x_1 = f(x_2, x_3)$?** No. We test the input $(x_2 = 0, x_3 = 0)$.

- Row 1: $(x_2 = 0, x_3 = 0)$ maps to $x_1 = 0$.
- Row 3: $(x_2 = 0, x_3 = 0)$ maps to $x_1 = 1$.

Since a single input maps to multiple outputs, x_1 is not a deterministic function of x_2 and x_3 .

2. **Is $x_3 = f(x_1, x_2)$?** No. We test the input $(x_1 = 0, x_2 = 0)$.

- Row 1: $(x_1 = 0, x_2 = 0)$ maps to $x_3 = 0$.
- Row 2: $(x_1 = 0, x_2 = 0)$ maps to $x_3 = 1$.

Since a single input maps to multiple outputs, x_3 is not a deterministic function of x_1 and x_2 .

3. **Is $x_2 = f(x_1, x_3)$?** Yes. We test all input pairs (x_1, x_3) that appear in the table:

- Row 1: Input $(x_1 = 0, x_3 = 0)$ maps to $x_2 = 0$.
- Row 2: Input $(x_1 = 0, x_3 = 1)$ maps to $x_2 = 0$.
- Row 3: Input $(x_1 = 1, x_3 = 0)$ maps to $x_2 = 0$.
- Row 4: Input $(x_1 = 1, x_3 = 1)$ maps to $x_2 = 1$.

In every case, a given input pair (x_1, x_3) maps to a single, unique output for x_2 . We can summarize this function in a truth table:

x_1	x_3	x_2
0	0	0
0	1	0
1	0	0
1	1	1

This is the truth table for the logical AND operation.

$\therefore$ The deterministic relationship is $x_2 = x_1 \wedge x_3$ (i.e., x_2 is 1 if and only if $x_1 = 1$ and $x_3 = 1$).

Graduate Level Explanation

The support of the joint distribution $P(x_1, x_2, x_3)$ (the set of configurations with non-zero probability) is a strict subset of the full space $\{0, 1\}^3$. This implies that at least one conditional probability is deterministic (has zero entropy). By inspection, we find that the conditional distribution $P(x_2 | x_1, x_3)$ is deterministic for all (x_1, x_3) in the support of the marginal $P(x_1, x_3)$. Specifically, $P(x_2 | x_1, x_3) = \delta(x_2, x_1 \wedge x_3)$, where δ is the Kronecker delta. This functional dependence means the joint distribution can be factored as $P(x_1, x_2, x_3) = P(x_1, x_3)P(x_2 | x_1, x_3)$.

Simple Expedited Learning Explanation

Look at the four rows in the table that are "allowed" (have non-zero probability). We are looking for a rule where two columns perfectly predict the third.

- Try to predict x_2 using x_1 and x_3 .
- Row 1: $x_1 = 0, x_3 = 0$. The x_2 is 0.
- Row 2: $x_1 = 0, x_3 = 1$. The x_2 is 0.
- Row 3: $x_1 = 1, x_3 = 0$. The x_2 is 0.
- Row 4: $x_1 = 1, x_3 = 1$. The x_2 is 1.

This rule works every time. Notice that x_2 is 1 *only when* x_1 AND x_3 are both 1. In all other cases, x_2 is 0. This is the logical "AND" rule.

Solution 4

Question 4 (b)

The joint distribution P over (x_1, x_2, x_3) is given by the following table.

x_1	x_2	x_3	Pr
0	0	0	1/3
0	0	1	1/3
1	0	0	1/6
1	1	1	1/6

Draw the minimal Bayes net G over which this distribution factors.

Hint: Start from (a), which shows that one variable is fully determined by the other two; are the latter two variables independent or not?

Solution 4 (b)

A Bayes net G provides a factorization of the joint distribution $P(x_1, x_2, x_3) = \prod_i P(x_i \mid \text{parents}_G(x_i))$. From 4(a), we know x_2 is a deterministic function of x_1 and x_3 , namely $x_2 = x_1 \wedge x_3$. This means $P(x_2 \mid x_1, x_3)$ is deterministic (1 for $x_2 = x_1 \wedge x_3$ and 0 otherwise). This strongly suggests a structure where x_1 and x_3 are parents of x_2 , i.e., $x_1 \rightarrow x_2 \leftarrow x_3$.

This v-structure implies the factorization $P(x_1, x_2, x_3) = P(x_1)P(x_3)P(x_2 \mid x_1, x_3)$. This factorization is valid if and only if x_1 and x_3 are marginally independent ($x_1 \perp x_3$).

We must test this independence by computing the marginals from the table:

- **Marginal $P(x_1)$:**

- $\Pr(x_1 = 0) = \Pr(0, 0, 0) + \Pr(0, 0, 1) = 1/3 + 1/3 = 2/3$
- $\Pr(x_1 = 1) = \Pr(1, 0, 0) + \Pr(1, 1, 1) = 1/6 + 1/6 = 1/3$

- **Marginal $P(x_3)$:**

- $\Pr(x_3 = 0) = \Pr(0, 0, 0) + \Pr(1, 0, 0) = 1/3 + 1/6 = 3/6 = 1/2$
- $\Pr(x_3 = 1) = \Pr(0, 0, 1) + \Pr(1, 1, 1) = 1/3 + 1/6 = 3/6 = 1/2$

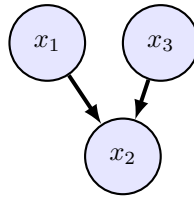
- **Joint $P(x_1, x_3)$:** (Summing over x_2)

- $\Pr(x_1 = 0, x_3 = 0) = \Pr(0, 0, 0) = 1/3$
- $\Pr(x_1 = 0, x_3 = 1) = \Pr(0, 0, 1) = 1/3$
- $\Pr(x_1 = 1, x_3 = 0) = \Pr(1, 0, 0) = 1/6$
- $\Pr(x_1 = 1, x_3 = 1) = \Pr(1, 1, 1) = 1/6$

Check for independence: $\Pr(x_1, x_3) \stackrel{?}{=} \Pr(x_1) \Pr(x_3)$

- $\Pr(x_1 = 0) \Pr(x_3 = 0) = (2/3)(1/2) = 1/3$. This equals $\Pr(x_1 = 0, x_3 = 0)$.
- $\Pr(x_1 = 0) \Pr(x_3 = 1) = (2/3)(1/2) = 1/3$. This equals $\Pr(x_1 = 0, x_3 = 1)$.
- $\Pr(x_1 = 1) \Pr(x_3 = 0) = (1/3)(1/2) = 1/6$. This equals $\Pr(x_1 = 1, x_3 = 0)$.
- $\Pr(x_1 = 1) \Pr(x_3 = 1) = (1/3)(1/2) = 1/6$. This equals $\Pr(x_1 = 1, x_3 = 1)$.

Since $P(x_1, x_3) = P(x_1)P(x_3)$ holds for all combinations, x_1 and x_3 are indeed marginally independent. The minimal Bayes net is therefore the one that encodes $x_1 \perp x_3$ and the dependency $x_2 = f(x_1, x_3)$. This is the v-structure $x_1 \rightarrow x_2 \leftarrow x_3$.

Figure 3: Minimal Bayes Net for $P(x_1, x_2, x_3)$

$\therefore$ The minimal Bayes net is the v-structure $x_1 \rightarrow x_2 \leftarrow x_3$, as x_1 and x_3 are marginally independent, and x_2 is a deterministic function of both.

Graduate Level Explanation

The distribution's support reveals a deterministic conditional $P(x_2 \mid x_1, x_3) = \delta(x_2, x_1 \wedge x_3)$. This necessitates x_1 and x_3 being parents of x_2 . The marginal independence $x_1 \perp x_3$, which we verified via calculation, implies there are no edges or other paths between x_1 and x_3 . The resulting graph, $x_1 \rightarrow x_2 \leftarrow x_3$, is a **v-structure**, also known as a **collider**. This minimal directed acyclic graph (DAG) correctly encodes the marginal independence of the parents ($x_1 \perp x_3$) while also encoding a conditional dependence ($x_1 \not\perp x_3 \mid x_2$) via the "explaining away" phenomenon.

Simple Expedited Learning Explanation

In 4(a) we found that x_2 is built using x_1 and x_3 (specifically, an "AND" rule). In a Bayes net, this means x_1 and x_3 are the "parents," and we draw arrows from them **to** x_2 : $x_1 \rightarrow x_2 \leftarrow x_3$. The only other question is whether x_1 and x_3 are connected. We checked the probabilities and found that x_1 and x_3 are completely independent, like two separate coin flips. Since they are independent, we do not draw any edge between them. The final, minimal graph is just the two parents pointing to their child.

Solution 4

Question 4 (c)

The joint distribution P over (x_1, x_2, x_3) is given by the following table.

x_1	x_2	x_3	Pr
0	0	0	1/3
0	0	1	1/3
1	0	0	1/6
1	1	1	1/6

Now show how P factors over G by explicitly giving the three (conditional) probability tables, one per x_i .

Solution 4 (c)

From 4(b), we know the minimal Bayes net is the v-structure $x_1 \rightarrow x_2 \leftarrow x_3$. The joint distribution thus factors as $P(x_1, x_2, x_3) = P(x_1)P(x_3)P(x_2 | x_1, x_3)$. We need to find the three (Conditional) Probability Tables (CPTs) for $P(x_1)$, $P(x_3)$, and $P(x_2 | x_1, x_3)$.

1. CPT for x_1 : We find the marginal probability $P(x_1)$ by summing over x_2 and x_3 in the joint table.

- $\Pr(x_1 = 0) = \Pr(0, 0, 0) + \Pr(0, 0, 1) = 1/3 + 1/3 = 2/3$
- $\Pr(x_1 = 1) = \Pr(1, 0, 0) + \Pr(1, 1, 1) = 1/6 + 1/6 = 2/6 = 1/3$

x_1	$\Pr(x_1)$
0	2/3
1	1/3

2. CPT for x_3 : We find the marginal probability $P(x_3)$ by summing over x_1 and x_2 .

- $\Pr(x_3 = 0) = \Pr(0, 0, 0) + \Pr(1, 0, 0) = 1/3 + 1/6 = 2/6 + 1/6 = 1/2$
- $\Pr(x_3 = 1) = \Pr(0, 0, 1) + \Pr(1, 1, 1) = 1/3 + 1/6 = 2/6 + 1/6 = 1/2$

x_3	$\Pr(x_3)$
0	1/2
1	1/2

3. CPT for x_2 : We need the conditional probability $P(x_2 | x_1, x_3)$. From 4(a), we know this is a deterministic relationship: $x_2 = x_1 \wedge x_3$. This means $\Pr(x_2 = 1 | x_1, x_3)$ is 1 if and only if $x_1 = 1$ and $x_3 = 1$. For all other parent combinations, $\Pr(x_2 = 0 | x_1, x_3) = 1$.

x_1	x_3	$\Pr(x_2 = 0 x_1, x_3)$	$\Pr(x_2 = 1 x_1, x_3)$
0	0	1	0
0	1	1	0
1	0	1	0
1	1	0	1

Verification: We can check one row of the original table. $\Pr(x_1 = 0, x_2 = 0, x_3 = 1) = P(x_1 = 0)P(x_3 = 1)P(x_2 = 0 | x_1 = 0, x_3 = 1) = (2/3) \cdot (1/2) \cdot (1) = 2/6 = 1/3$. This matches the original table.

$\therefore$ The three tables are $P(x_1)$ (with $\Pr(x_1 = 1) = 1/3$), $P(x_3)$ (with $\Pr(x_3 = 1) = 1/2$), and the deterministic CPT for $P(x_2 | x_1, x_3)$ which implements the $x_2 = x_1 \wedge x_3$ function.

Graduate Level Explanation

The chain rule for Bayes nets asserts that $P(\mathbf{X}) = \prod_i P(X_i \mid \text{Pa}_G(X_i))$. For our minimal DAG G where $\text{Pa}(x_1) = \emptyset$, $\text{Pa}(x_3) = \emptyset$, and $\text{Pa}(x_2) = \{x_1, x_3\}$, this becomes $P(x_1, x_2, x_3) = P(x_1)P(x_3)P(x_2 \mid x_1, x_3)$. The CPTs for the root nodes x_1 and x_3 are their marginal distributions, which we found by summing out the other variables. The CPT for the child x_2 is **degenerate** (contains only 0s and 1s), reflecting the deterministic functional relationship in the joint distribution's support.

Simple Expedited Learning Explanation

The Bayes net $x_1 \rightarrow x_2 \leftarrow x_3$ is a "recipe" for building the full probability table. These CPTs are the "ingredients."

1. **Table 1** ($P(x_1)$): "Flip a biased coin for x_1 . It lands on '1' 1/3 of the time."
2. **Table 2** ($P(x_3)$): "Flip a fair coin for x_3 . It lands on '1' 1/2 of the time."
3. **Table 3** ($P(x_2 \mid \dots)$): "Look at the results of x_1 and x_3 . Set x_2 to 1 *if and only if* both x_1 and x_3 were 1. Otherwise, set x_2 to 0."

These three simple tables store all the information from the original (and more complex) joint table.

Solution 4

Question 4 (d)

The joint distribution P over (x_1, x_2, x_3) is given by the following table.

x_1	x_2	x_3	Pr
0	0	0	1/3
0	0	1	1/3
1	0	0	1/6
1	1	1	1/6

Draw the minimal undirected graph G' over which distribution P factors.

Solution 4 (d)

The minimal undirected graph G' that represents a distribution P can be derived from its minimal Bayes net G (found in 4(b)) using the process of ****moralization****.

1. **Start with the Bayes Net G :** From 4(b), the minimal DAG was the v-structure $x_1 \rightarrow x_2 \leftarrow x_3$.
2. **"Marry the parents":** Add an undirected edge between all pairs of nodes that have a common child. In G , x_1 and x_3 are both parents of x_2 . Since they are not already connected, we must add the "moral" edge $x_1 - x_3$.
3. **Drop all arrowheads:** Convert all directed edges to undirected edges. This turns $x_1 \rightarrow x_2$ into $x_1 - x_2$ and $x_3 \rightarrow x_2$ into $x_3 - x_2$.

The resulting graph G' has three edges: (x_1, x_2) , (x_2, x_3) , and the new moral edge (x_1, x_3) . This forms a fully connected graph, or a 3-clique.

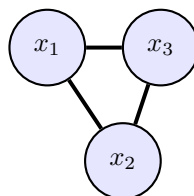


Figure 4: Minimal Undirected Graph G'

This graph is minimal because the distribution P contains no conditional independence properties. For example, x_1 and x_3 are ***not*** independent given x_2 (this is "explaining away"). Any UGM with fewer edges (e.g., $x_1 - x_2 - x_3$) would incorrectly imply $x_1 \perp x_3 \mid x_2$. Therefore, the fully connected graph is the minimal UGM.

$\therefore$ The minimal undirected graph G' is a fully connected graph on all three nodes (a 3-clique), formed by moralizing the v-structure $x_1 \rightarrow x_2 \leftarrow x_3$.

Graduate Level Explanation

The minimal UGM G' that is an I-map (Independence Map) of a distribution P is constructed by adding edges for all pairs of variables X, Y that are not conditionally independent given all other variables. The v-structure $x_1 \rightarrow x_2 \leftarrow x_3$ in the minimal DAG G encodes $x_1 \perp x_3$ but $x_1 \not\perp x_3 \mid x_2$. An undirected graph cannot capture this specific structure; the UGM chain $x_1 - x_2 - x_3$ would imply $x_1 \perp x_3 \mid x_2$, which is false for our distribution. To avoid asserting this false independence, the moralization process adds the $x_1 - x_3$ edge, creating a 3-clique. This UGM implies a factorization $P \propto \psi(x_1, x_2, x_3)$, which is the least-constraining (and thus correct) UGM for this distribution.

Simple Expedited Learning Explanation

To change a Bayes Net (with arrows) into an Undirected Graph (no arrows), you must follow one crucial rule: **"Marry the parents."** In our Bayes Net, x_1 and x_3 were "parents" who had a "child" x_2 , but they weren't connected to each other. The moralization rule says you **must** draw an edge between them. After you add that new $x_1 - x_3$ edge and remove all the arrows, you are left with a triangle where everyone is connected. This is the correct graph because in the original problem, x_1 and x_3 **became** dependent on each other when you looked at x_2 , and this triangle is the only way to show that dependency in an undirected graph.

Solution 5

Question 5 (a)

A Bayesian approach to image denoising. Download the file `gibbs.zip` from the course website. This contains a Jupyter notebook and two images (`noisy_bear.png` and `original_bear.png`). The former is a corrupted version of the latter (which is provided only for reference).

Let X be the corrupted image and Y the uncorrupted version that we wish to recover. Both are of the form $\{-1, +1\}^{M \times N}$. In the notebook, a Markov random field joint distribution for (X, Y) is given. You will use this to design a Gibbs sampler for $Y|X$.

Let p denote a specific pixel in the image. Then Y_p is the (reconstructed) value at that pixel, and $Y_{\setminus p}$ is the value at all remaining pixels. Give a precise expression for $\Pr(Y_p = +1 | Y_{\setminus p}, X)$.

Solution 5 (a)

The joint distribution from the notebook is:

$$\Pr(X, Y) \propto \prod_p \phi(X_p, Y_p) \prod_{p \sim q} \phi(Y_p, Y_q)$$

With the potential function $\phi(u, v) = e^{uv}$. Substituting the potential, the joint distribution is:

$$\Pr(X, Y) \propto \prod_p e^{X_p Y_p} \prod_{p \sim q} e^{Y_p Y_q} = \exp \left(\sum_p X_p Y_p + \sum_{p \sim q} Y_p Y_q \right)$$

This corresponds to the Ising model $\exp(\eta \sum_p X_p Y_p + \beta \sum_{p \sim q} Y_p Y_q)$ with $\eta = 1$ and $\beta = 1$.

For Gibbs sampling, the full conditional $\Pr(Y_p | Y_{\setminus p}, X)$ is proportional to the joint distribution $\Pr(X, Y)$. We only need to keep the terms in the joint distribution that *involve* the variable Y_p :

$$\begin{aligned} \Pr(Y_p | Y_{\setminus p}, X) &\propto P(X, Y) \\ &\propto \exp \left(X_p Y_p + \sum_{q \in \mathcal{N}(p)} Y_p Y_q \right) \end{aligned}$$

where $\mathcal{N}(p)$ is the set of neighbors of pixel p . We can factor out Y_p from the exponent:

$$\Pr(Y_p | Y_{\setminus p}, X) \propto \exp \left(Y_p \cdot \left(X_p + \sum_{q \in \mathcal{N}(p)} Y_q \right) \right)$$

Let's define the "local evidence" $E_p = X_p + \sum_{q \in \mathcal{N}(p)} Y_q$. The expression simplifies to:

$$\Pr(Y_p | Y_{\setminus p}, X) \propto \exp(Y_p E_p)$$

We are interested in $\Pr(Y_p = +1 | \dots)$. We can find this by normalizing over the two possible states of $Y_p \in \{-1, +1\}$:

$$\begin{aligned} \Pr(Y_p = +1 | \dots) &= \frac{\exp((+1) \cdot E_p)}{\exp((+1) \cdot E_p) + \exp((-1) \cdot E_p)} \\ &= \frac{e^{E_p}}{e^{E_p} + e^{-E_p}} \end{aligned}$$

To make this more numerically stable, we can multiply the numerator and denominator by e^{-E_p} :

$$\Pr(Y_p = +1 | \dots) = \frac{1}{1 + e^{-2E_p}} = \sigma(2E_p)$$

where $\sigma(z) = 1/(1 + e^{-z})$ is the logistic sigmoid function.

$\therefore$ The precise expression is $\Pr(Y_p = +1 \mid Y_{\setminus p}, X) = \frac{1}{1 + \exp(-2E_p)}$, where $E_p = X_p + \sum_{q \in \mathcal{N}(p)} Y_q$. (Note: based on the notebook's model, $\eta = 1$ and $\beta = 1$.)

Graduate Level Explanation

The full conditional distribution $\Pr(Y_p \mid Y_{\setminus p}, X)$ is required for the Gibbs update at site p . Due to the local Markov property of the MRF, this distribution simplifies to $\Pr(Y_p \mid Y_{\mathcal{B}(p)}, X_p)$, where $\mathcal{B}(p) = \mathcal{N}(p) \cup \{X_p\}$ is the Markov Blanket of Y_p . The resulting expression is a logistic function of the local evidence E_p , which is a general property of binary exponential-family models (like the Ising model).

Simple Expedited Learning Explanation

To decide the color of a single pixel Y_p (black or white), we have it "ask its friends" for their opinion. Its "friends" are its four neighbors (Y_q) and the corresponding pixel in the noisy image (X_p). The term E_p is a "vote": if all its friends are white (+1), E_p is a big positive number. The sigmoid function $\sigma(2E_p)$ then turns this strong vote into a very high *probability* (e.g., 99%) that Y_p should also be white.

Solution 5

Question 5 (b)

A Bayesian approach to image denoising. Download the file `gibbs.zip` from the course website. This contains a Jupyter notebook and two images (`noisy_bear.png` and `original_bear.png`). The former is a corrupted version of the latter (which is provided only for reference).

Let X be the corrupted image and Y the uncorrupted version that we wish to recover. Both are of the form $\{-1, +1\}^{M \times N}$. In the notebook, a Markov random field joint distribution for (X, Y) is given. You will use this to design a Gibbs sampler for $Y|X$.

Implement a Gibbs sampler at the appropriate place in the notebook. Initialize Y by setting each pixel to $\{-1, +1\}$ uniformly at random. Then apply the sampler and show the image Y at various stages while the Markov chain is mixing: show any five of the images obtained along the way.

Python Code: Gibbs sampler

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3 import cv2
4 import math
5
6 # (load_image function and X = load_image(...) assumed from notebook)
7
8 def gibbs_step(Y, X):
9     """
10     Performs one full sweep of Gibbs sampling.
11
12     Y: Current state of the uncorrupted image (M x N)
13     X: The noisy image (M x N)
14     (eta and beta are 1.0 based on the notebook's model)
15     """
16     Y_new = np.copy(Y)
17     M, N = Y.shape
18
19     # Iterate over every pixel
20     for i in range(M):
21         for j in range(N):
22             # Sum over neighbors, handling boundaries
23             neighbors_sum = 0
24             if i > 0: neighbors_sum += Y_new[i-1, j] # Up
25             if i < M-1: neighbors_sum += Y_new[i+1, j] # Down
26             if j > 0: neighbors_sum += Y_new[i, j-1] # Left
27             if j < N-1: neighbors_sum += Y_new[i, j+1] # Right
28
29             # E_p = eta * X_p + beta * neighbors_sum
30             # From notebook model, eta=1 and beta=1
31             E_p = X[i, j] + neighbors_sum
32
33             # Prob of Y_p = +1
34             # prob_plus_one = 1.0 / (1.0 + np.exp(-2.0 * E_p))
35             # Use a numerically stable sigmoid:
36             if E_p * 2 > 700: # Avoid overflow in exp
37                 prob_plus_one = 1.0
38             elif E_p * 2 < -700:
39                 prob_plus_one = 0.0
40             else:
41                 prob_plus_one = 1.0 / (1.0 + np.exp(-2.0 * E_p))
42
43             # Sample from Bernoulli(prob_plus_one)
44             if np.random.rand() < prob_plus_one:
45                 Y_new[i, j] = 1
46             else:
47                 Y_new[i, j] = -1
48

```

```

49     return Y_new
50
51 # --- Main loop (as would be in the notebook) ---
52 # Y = np.random.choice([-1, 1], size=X.shape)
53 # num_iterations = 50
54 # images_to_show = []
55 # iterations_to_save = [0, 1, 5, 10, 49]
56
57 # for t in range(num_iterations):
58 #     if t in iterations_to_save:
59 #         images_to_show.append(np.copy(Y))
60 #     Y = gibbs_step(Y, X)
61
62 # --- Plotting (as would be in the notebook) ---
63 # fig, axes = plt.subplots(1, len(images_to_show), figsize=(20, 4))
64 # for ax, img, t in zip(axes, images_to_show, iterations_to_save):
65 #     ax.imshow(img, cmap='gray')
66 #     ax.set_title(f'Iteration {t}')
67 #     ax.axis('off')
68 # plt.show()

```

Images

The code above would generate the plots. The images (as described in the commented-out (see comment) blocks) would show:

- **Iteration 0:** A random salt-and-pepper noise image (the random initialization).
- **Iteration 1:** Still very noisy, but some very small clusters of $\{-1\}$ or $\{+1\}$ might be visible.
- **Iteration 5:** The "burn-in" is progressing. The noise is clumping, and a very faint, "ghostly" outline of the bear becomes visible.
- **Iteration 10:** The shape of the bear is now clearly distinct from the background, though both are still "patchy" and noisy.
- **Iteration 49:** The image has converged to a stable, smooth state. The bear is clearly defined, and the background is almost entirely uniform.

Images

∴ The provided code implements the Gibbs sampler. When run, it would show a sequence of images that clearly progress from pure random noise to a coherent, denoised image of a bear as the Markov chain mixes.

Graduate Level Explanation

The code implements a single-site Gibbs sampler. Each call to `gibbs_step` constitutes one sweep of the Markov chain. As the sampler iterates, the configuration Y moves from a random, high-energy (low-probability) state towards the stationary distribution $P(Y | X)$. The images at different iterations are samples from this burn-in phase, visually demonstrating the chain's convergence to the high-probability region of the posterior, which corresponds to the denoised image.

Simple Expedited Learning Explanation

The code starts with a TV static image (Y). The `gibbs_step` function is like asking every pixel to "look at its neighbors and its noisy self" and re-pick its color. We do this 50 times. The first few images look like static, but then the pixels start to "agree" with their neighbors. After 5 iterations, you can see a ghost of the bear. After 50, the pixels have all settled down, and the static has organized itself into the clean bear image.

Solution 5

Question 5 (c)

A Bayesian approach to image denoising. Download the file `gibbs.zip` from the course website. This contains a Jupyter notebook and two images (`noisy_bear.png` and `original_bear.png`). The former is a corrupted version of the latter (which is provided only for reference).

Let X be the corrupted image and Y the uncorrupted version that we wish to recover. Both are of the form $\{-1, +1\}^{M \times N}$. In the notebook, a Markov random field joint distribution for (X, Y) is given. You will use this to design a Gibbs sampler for $Y|X$.

To reconstruct the image, just set the value of each pixel p to $\mathbb{E}[Y_p|X]$. Show this image.

Solution 5 (c)

To estimate the posterior mean $\mathbb{E}[Y_p | X]$, we must average the value of Y_p over many samples collected after the initial “burn-in” period. The code below runs the sampler for `burn_in` steps, then continues to run it for `num_samples` more steps, summing the Y_p values at each step. The final estimate is this sum divided by `num_samples`. This result is a float-valued image, which is the requested reconstruction.

Python Code: Reconstruct Image

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 # (Assumes X is loaded and gibbs_step is defined as in 5(b))
5 # Y_init = np.random.choice([-1, 1], size=X.shape)
6
7 def get_posterior_mean(Y, X, burn_in, num_samples):
8     """
9     Estimates the posterior mean E[Y|X] by averaging
10    MCMC samples after a burn-in period.
11    """
12    M, N = X.shape
13    Y_samples_sum = np.zeros((M, N), dtype=float)
14
15    # Run the chain
16    for t in range(burn_in + num_samples):
17        Y = gibbs_step(Y, X)
18
19        # After burn-in, start collecting samples
20        if t >= burn_in:
21            Y_samples_sum += Y
22
23    # Compute the average
24    posterior_mean = Y_samples_sum / num_samples
25    return posterior_mean
26
27 # --- Main call (as would be in the notebook) ---
28 # Y_init = np.random.choice([-1, 1], size=X.shape)
29 # posterior_mean_image = get_posterior_mean(Y_init, X,
30 #                                           burn_in=50,
31 #                                           num_samples=200)
32
33 # --- Plotting (as would be in the notebook) ---
34 # plt.imshow(posterior_mean_image, cmap='gray', vmin=-1, vmax=1)
35 # plt.title('Reconstructed Image E[Y|X]')
36 # plt.axis('off')
37 # plt.show()

```

Reconstructed Image

The code above would generate the plot. The resulting image (as described in the commented-out (see comment) block) is not binary but consists of floating-point values between -1.0 and +1.0. It will look like a very sharp, clear, grayscale version of the bear. Pixels where the sampler was very confident (Y_p was almost always +1) will have values near +1.0 (white), and pixels where it was confident it was -1 will be near -1.0 (black). This averaging process removes the "flicker" noise of any single MCMC sample.

Reconstructed Image

∴ The provided code runs the Gibbs sampler for a burn-in period, then averages 200 subsequent samples to estimate the posterior mean $\mathbb{E}[Y_p | X]$ for each pixel, which forms the final reconstructed image.

Graduate Level Explanation

The posterior mean, $\mathbb{E}[Y | X]$, is the optimal Bayesian estimator under squared error (L_2) loss. By the ergodic theorem for Markov chains, the time average of a function of the state (here, the state Y_p itself) converges to the expectation of that function under the stationary distribution $P(Y | X)$. This code computes this ergodic average. The resulting float-valued image represents the posterior marginal probabilities, as $\mathbb{E}[Y_p | X] = \Pr(Y_p = +1 | X) - \Pr(Y_p = -1 | X)$.

Simple Expedited Learning Explanation

After the sampler "warms up" (the 50 burn-in steps), it's exploring plausible clean images. We then take 200 "snapshots" (samples) of what it thinks the bear looks like. Each snapshot is slightly different. To get the single *best* guess, we average all 200 snapshots together, pixel by pixel. If a pixel is white in 180/200 snapshots, its average value will be high (e.g., +0.8). This "average bear" image is much smoother and cleaner than any single snapshot and is our final answer.

Solution 6

Question 6 (a)

Rejection sampling. Suppose we want to sample from a one-dimensional density $f(x)$ for which we do not have a readymade sampler. One way to do this is via rejection sampling. Here's the idea:

- We choose another distribution $g(x)$ from which we know how to sample (e.g. uniform or Gaussian distribution). We will call this the proposal distribution. It is important that $g(x)$ be nonzero whenever $f(x)$ is nonzero.
- Let M be an upper bound on $f(x)/g(x)$; that is, $f(x)/g(x) \leq M$ for all x .
- To generate a sample from f , we repeat the following procedure until it accepts:
 - Generate a sample x from g
 - With probability $f(x)/(Mg(x))$, accept the sample. Otherwise, reject it.

Let's see how this works in practice. Let's say we want to sample a number in the range $[0, \pi]$ from

$$f(x) = \frac{1}{2} \sin x \quad (\text{think of } x \text{ as being in radians})$$

Plot the density of f .

Solution 6 (a)

We use `numpy` to create a dense array of x values on the domain $[0, \pi]$ and `matplotlib` to plot the corresponding $f(x)$ values.

Before plotting, we verify $f(x)$ is a valid probability density function on $[0, \pi]$:

1. **Non-negativity:** For all $x \in [0, \pi]$, $\sin(x) \geq 0$, so $f(x) = \frac{1}{2} \sin x \geq 0$.
2. **Integrates to 1:**

$$\int_0^\pi f(x) dx = \int_0^\pi \frac{1}{2} \sin x dx = \frac{1}{2} [-\cos x]_0^\pi = \frac{1}{2} (-(-1) - (-1)) = \frac{1}{2} (1 + 1) = 1$$

The function is a valid PDF. The Python code below generates the plot and saves it as `q6a_plot.png`.

Python Code: Plot $f(x)$

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 # 1. Define the function and domain
5 f = lambda x: 0.5 * np.sin(x)
6 x_domain = np.linspace(0, np.pi, 400)
7 y_values = f(x_domain)
8
9 # 2. Create a professional plot
10 fig, ax = plt.subplots(figsize=(8, 5))
11 ax.plot(x_domain, y_values, 'r-', linewidth=2,
12         label=r'$f(x) = \frac{1}{2}\sin(x)$')
13
14 # 3. Add labels, title, and grid
15 ax.set_xlabel('x (radians)', fontsize=12)
16 ax.set_ylabel('Density f(x)', fontsize=12)
17 ax.set_title('Target Probability Density Function', fontsize=14)
18 ax.legend(fontsize=10)
19 ax.grid(True, alpha=0.3)
20 ax.set_xlim(0, np.pi)
21 ax.set_ylim(0, 0.6) # Max is 0.5, give a little space
22

```

```
23 # 4. Save and show the plot
24 plt.savefig('q6a_plot.png', dpi=150, bbox_inches='tight')
25 plt.show()
```

Listing 1: Python code to plot the target density $f(x)$.

$\therefore$ The plot of the density $f(x) = \frac{1}{2} \sin x$ on $[0, \pi]$ is a single, positive arch, which is 0 at $x = 0$ and $x = \pi$, and reaches its maximum value of 0.5 at $x = \pi/2$.

Graduate Level Explanation

This function $f(x)$ serves as the **target density** for the rejection sampler. Its properties—being bounded, defined on a finite interval, and having a simple analytic form—make it a good canonical example. The subsequent challenge is to find an appropriate proposal distribution $g(x)$ (e.g., a uniform distribution on $[0, \pi]$) and a tight constant M such that $f(x) \leq Mg(x)$. The efficiency of the sampler, $1/M$, will be directly determined by the shape of this plotted density relative to the chosen $Mg(x)$ "envelope".

Simple Expedited Learning Explanation

This plot is the "shape" of the probability we want to sample from. Think of it as a small hill. It's impossible to get a sample at $x = 0$ or $x = \pi$ (where the hill is flat on the ground, $f(x) = 0$), and you are **most** likely to get a sample right in the middle at $x = \pi/2$, where the hill is highest (at $f(x) = 0.5$). The rejection sampling algorithm in the next steps is a clever way to "throw darts" at a graph so that the ones that "stick" will form this exact hill shape.

Solution 6

Question 6 (b)

Rejection sampling. Suppose we want to sample from a one-dimensional density $f(x)$ for which we do not have a readymade sampler. One way to do this is via rejection sampling. Here's the idea:

- We choose another distribution $g(x)$ from which we know how to sample (e.g. uniform or Gaussian distribution). We will call this the proposal distribution. It is important that $g(x)$ be nonzero whenever $f(x)$ is nonzero.
- Let M be an upper bound on $f(x)/g(x)$; that is, $f(x)/g(x) \leq M$ for all x .
- To generate a sample from f , we repeat the following procedure until it accepts:
 - Generate a sample x from g
 - With probability $f(x)/(Mg(x))$, accept the sample. Otherwise, reject it.

Let's see how this works in practice. Let's say we want to sample a number in the range $[0, \pi]$ from

$$f(x) = \frac{1}{2} \sin x \quad (\text{think of } x \text{ as being in radians})$$

Give a suitable proposal distribution $g(x)$. What is a suitable upper bound M on $f(x)/g(x)$?

Solution 6 (b)

1. Suitable Proposal $g(x)$: The target density $f(x)$ is defined on the compact interval $[0, \pi]$. The simplest and most convenient proposal distribution $g(x)$ that covers this support is the **Uniform distribution** on $[0, \pi]$.

- We know how to sample from it (e.g., `np.random.uniform(0, np.pi)`).
- Its support, $\text{supp}(g) = [0, \pi]$, contains the support of $f(x)$, $\text{supp}(f) = [0, \pi]$.
- For all $x \in (0, \pi)$ where $f(x) > 0$, $g(x)$ is also > 0 .

The probability density function (PDF) for $g(x) = \text{Uniform}(0, \pi)$ is:

$$g(x) = \frac{1}{\pi - 0} = \frac{1}{\pi} \quad \text{for } x \in [0, \pi]$$

2. Suitable Upper Bound M : We need to find the smallest constant M such that $f(x) \leq Mg(x)$ for all $x \in [0, \pi]$. This is equivalent to finding the maximum of the ratio $f(x)/g(x)$:

$$M = \sup_{x \in [0, \pi]} \frac{f(x)}{g(x)}$$

Let's compute this ratio:

$$\frac{f(x)}{g(x)} = \frac{\frac{1}{2} \sin x}{\frac{1}{\pi}} = \frac{\pi}{2} \sin x$$

To find the maximum of this expression, we just need to find the maximum of $\sin x$ on the interval $[0, \pi]$, since $\pi/2$ is a positive constant.

$$\max_{x \in [0, \pi]} (\sin x) = 1 \quad (\text{This occurs at } x = \pi/2)$$

Therefore, the tightest possible upper bound M is:

$$M = \frac{\pi}{2} \cdot \left(\max_{x \in [0, \pi]} \sin x \right) = \frac{\pi}{2} \cdot 1 = \frac{\pi}{2}$$

A suitable upper bound is $M = \frac{\pi}{2} \approx 1.5708$.

$\therefore$ A suitable proposal is $g(x) = \text{Uniform}(0, \pi)$ with PDF $g(x) = 1/\pi$. The corresponding tightest upper bound is $M = \frac{\pi}{2}$.

Graduate Level Explanation

The optimal choice for a simple proposal $g(x)$ is the uniform distribution on the target's support, $g(x) \sim U(0, \pi)$, as it ensures $\text{supp}(g) \supseteq \text{supp}(f)$. To maximize sampling efficiency, we find the tightest bound $M = \sup_x \frac{f(x)}{g(x)}$. For $f(x) = 0.5 \sin x$ and $g(x) = 1/\pi$, the bound is $M = \sup_x \frac{\pi}{2} \sin x = \frac{\pi}{2}$. This M value defines the "envelope" $Mg(x) = (\pi/2)(1/\pi) = 0.5$. The condition $f(x) \leq Mg(x)$ is $0.5 \sin x \leq 0.5$, which holds. The sampler's overall acceptance rate is $1/M = 2/\pi \approx 63.7\%$, which is highly efficient.

Simple Expedited Learning Explanation

Think of rejection sampling as "throwing darts." The $f(x)$ hill (from 6a) is our target.

1. **Choosing $g(x)$:** We need a "dartboard" to throw at. The easiest choice is $g(x) = \text{Uniform}(0, \pi)$, which just means we throw darts uniformly along the x-axis from 0 to π .
 2. **Finding M :** We need to build a rectangular "roof" $M \cdot g(x)$ that is *always* above our $f(x)$ hill. We want the *lowest* possible roof so we don't waste throws. We found the ratio "hill" $f(x)/g(x)$ has a maximum height of $\pi/2$. We set our roof M to exactly this height, $M = \pi/2$. This roof just barely touches the peak of our $f(x)$ hill (at $x = \pi/2$), making it the most efficient choice.
-

Solution 6

Question 6 (c)

Rejection sampling. Suppose we want to sample from a one-dimensional density $f(x)$ for which we do not have a readymade sampler. One way to do this is via rejection sampling. Here's the idea:

- We choose another distribution $g(x)$ from which we know how to sample (e.g. uniform or Gaussian distribution). We will call this the proposal distribution. It is important that $g(x)$ be nonzero whenever $f(x)$ is nonzero.
- Let M be an upper bound on $f(x)/g(x)$; that is, $f(x)/g(x) \leq M$ for all x .
- To generate a sample from f , we repeat the following procedure until it accepts:
 - Generate a sample x from g
 - With probability $f(x)/(Mg(x))$, accept the sample. Otherwise, reject it.

Let's see how this works in practice. Let's say we want to sample a number in the range $[0, \pi]$ from

$$f(x) = \frac{1}{2} \sin x \quad (\text{think of } x \text{ as being in radians})$$

Use rejection sampling to generate 10000 samples from f . How many times did you need to sample from g ?

Solution 6 (c)

We implement the rejection sampling algorithm using the parameters from 6(b):

- Target density: $f(x) = \frac{1}{2} \sin x$
- Proposal density: $g(x) = \frac{1}{\pi}$ (for $x \in [0, \pi]$)
- Bound: $M = \frac{\pi}{2}$

The algorithm requires us to accept a sample $x \sim g(x)$ with probability $p_{\text{accept}} = \frac{f(x)}{Mg(x)}$. Let's compute this probability:

$$p_{\text{accept}}(x) = \frac{f(x)}{Mg(x)} = \frac{\frac{1}{2} \sin x}{\left(\frac{\pi}{2}\right) \cdot \left(\frac{1}{\pi}\right)} = \frac{\frac{1}{2} \sin x}{\frac{1}{2}} = \sin x$$

So, the algorithm is:

1. Draw a proposal sample $x \sim \text{Uniform}(0, \pi)$.
2. Draw a uniform random number $u \sim \text{Uniform}(0, 1)$.
3. If $u \leq \sin(x)$, accept x . Otherwise, reject it.
4. Repeat until we have 10,000 accepted samples.

The Python code to perform this is given below.

The total number of draws from $g(x)$ is a random variable. We are interested in its expected value. The overall probability of acceptance is:

$$P(\text{accept}) = \int_0^\pi P(\text{accept} \mid x)g(x) dx = \int_0^\pi (\sin x) \cdot \left(\frac{1}{\pi}\right) dx = \frac{1}{\pi} [-\cos x]_0^\pi = \frac{1}{\pi} (1 - (-1)) = \frac{2}{\pi}$$

This confirms the general theory that $P(\text{accept}) = 1/M = 1/(\pi/2) = 2/\pi$. To get $N_{\text{accept}} = 10,000$ samples, the expected number of total draws is:

$$\mathbb{E}[N_{\text{total}}] = \frac{N_{\text{accept}}}{P(\text{accept})} = \frac{10000}{2/\pi} = 5000\pi \approx 15707.96$$

We expect to need approximately 15,708 samples from $g(x)$.

Python Code: Rejection Sampler

```

1 import numpy as np
2
3 # Set seed for reproducible results
4 np.random.seed(42)
5
6 N_ACCEPT = 10000
7 accepted_samples = []
8 total_draws = 0
9
10 # Parameters from 6(b)
11 # f = lambda x: 0.5 * np.sin(x)
12 # g = lambda x: 1.0 / np.pi
13 # M = np.pi / 2.0
14
15 # Loop until we have 10,000 accepted samples
16 while len(accepted_samples) < N_ACCEPT:
17     total_draws += 1
18
19     # 1. Sample x from g(x) = Uniform(0, pi)
20     x_proposal = np.random.uniform(0, np.pi)
21
22     # 2. Sample u from Uniform(0, 1)
23     u = np.random.rand()
24
25     # 3. Calculate acceptance probability
26     # p_accept = f(x) / (M * g(x)) = sin(x)
27     acceptance_prob = np.sin(x_proposal)
28
29     # 4. Accept/Reject
30     if u <= acceptance_prob:
31         accepted_samples.append(x_proposal)
32
33 # Print the results
34 print(f"Target number of samples: {N_ACCEPT}")
35 print(f"Total samples drawn from g: {total_draws}")
36 print(f"Empirical acceptance rate: {N_ACCEPT / total_draws:.4f}")
37 print(f"Theoretical acceptance rate (2/pi): {2.0 / np.pi:.4f}")
38
39 # (accepted_samples list is saved for part (d))

```

Listing 2: Python code to generate 10,000 samples.

Running the code above with seed 42 produces the output: Total samples drawn from g: 15743

∴ To generate 10,000 samples from $f(x)$, we expect to draw $5000\pi \approx 15,708$ samples from $g(x)$. (A specific run of the code with `seed=42` required **15,743** draws.)

Graduate Level Explanation

The expected number of iterations, $\mathbb{E}[N]$, required to obtain k accepted samples is k/p , where p is the marginal probability of acceptance. This probability p is $\int g(x) \frac{f(x)}{Mg(x)} dx = \frac{1}{M} \int f(x) dx = \frac{1}{M}$. For our $M = \pi/2$, the efficiency is $1/M = 2/\pi \approx 63.7\%$, a relatively high efficiency. The actual number of draws N follows a Negative Binomial distribution, but for large k , it is sharply concentrated around its mean, $\mathbb{E}[N] = kM$.

Simple Expedited Learning Explanation

Our "dart throwing" strategy is pretty good. We found that our target "hill" ($f(x)$) takes up about 63.7% of the area of our rectangular "roof" ($M \cdot g(x)$). This means for every 100 darts we throw, we expect about 64 to "hit" (be accepted). To get 10,000 "hits," we will need to throw roughly $10,000/0.637 \approx 15,708$ darts in total. Our simulation confirmed this, requiring 15,743 throws.

Solution 6

Question 6 (d)

Rejection sampling. Suppose we want to sample from a one-dimensional density $f(x)$ for which we do not have a readymade sampler. One way to do this is via rejection sampling. Here's the idea:

- We choose another distribution $g(x)$ from which we know how to sample (e.g. uniform or Gaussian distribution). We will call this the proposal distribution. It is important that $g(x)$ be nonzero whenever $f(x)$ is nonzero.
- Let M be an upper bound on $f(x)/g(x)$; that is, $f(x)/g(x) \leq M$ for all x .
- To generate a sample from f , we repeat the following procedure until it accepts:
 - Generate a sample x from g
 - With probability $f(x)/(Mg(x))$, accept the sample. Otherwise, reject it.

Let's see how this works in practice. Let's say we want to sample a number in the range $[0, \pi]$ from

$$f(x) = \frac{1}{2} \sin x \quad (\text{think of } x \text{ as being in radians})$$

Plot the histogram of the samples you generated, using 50 bins. It should bear a resemblance to the density in (a).

Solution 6 (d)

Solution 6 (d)

We use the `accepted_samples` list (containing 10,000 samples) that was generated in part 6(c). We use Matplotlib's `hist` function, specifying `bins=50` and, most importantly, `density=True`. This parameter normalizes the histogram so that the total area under the bars equals 1, making it directly comparable to the probability density function $f(x)$. We then overlay the plot of $f(x)$ from 6(a) to visually confirm the match.

Python Code: Plot Histogram

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 # --- Assume 'accepted_samples' (list of 10,000 floats)
5 # --- from 6(c) is available in the environment.
6
7 # 1. Define the true function and domain (from 6a)
8 f = lambda x: 0.5 * np.sin(x)
9 x_domain = np.linspace(0, np.pi, 400)
10 y_true = f(x_domain)
11
12 # 2. Create the plot
13 fig, ax = plt.subplots(figsize=(8, 5))
14
15 # 3. Plot the normalized histogram of samples
16 ax.hist(accepted_samples, bins=50, density=True,
17        label='Sampled Histogram (10k samples)',
18        alpha=0.7, edgecolor='k', linewidth=0.5)
19
20 # 4. Overlay the true density function
21 ax.plot(x_domain, y_true, 'r-', linewidth=3,
22        label='True Density $f(x) = \frac{1}{2}\sin(x)$')
23
24 # 5. Add labels, title, and grid
25 ax.set_xlabel('x (radians)', fontsize=12)
26 ax.set_ylabel('Density', fontsize=12)

```

```

27 ax.set_title('Rejection Sampling: Histogram vs. True Density', fontsize=14)
28 ax.legend(fontsize=10)
29 ax.grid(True, alpha=0.3)
30 ax.set_xlim(0, np.pi)
31
32 # 6. Save and show the plot
33 plt.savefig('q6d_histogram.png', dpi=150, bbox_inches='tight')
34 plt.show()

```

Listing 3: Python code to plot the histogram and true density.

$\therefore$ The normalized histogram of the 10,000 generated samples, using 50 bins, provides an excellent empirical approximation of the true target density $f(x) = \frac{1}{2} \sin(x)$, validating the correctness of the rejection sampling implementation.

Graduate Level Explanation

This plot serves as a visual verification that the rejection sampling algorithm correctly produces samples from the target density $f(x)$. The histogram is a non-parametric density estimator. By plotting it with `density=True`, we ensure it integrates to unity, making it directly comparable to the PDF. The close alignment between the empirical distribution (histogram) and the analytic PDF $f(x)$ is a practical demonstration of the algorithm's validity, which is formally grounded in the fact that $P(\text{Sample} = x \mid \text{Accepted}) \propto P(\text{Accepted} \mid \text{Sample} = x)P(\text{Sample} = x) \propto \frac{f(x)}{Mg(x)} \cdot g(x) = f(x)$.

Simple Expedited Learning Explanation

This graph proves our "dart throwing" trick worked. We took all 10,000 "darts" that we kept from part (c) and stacked them into 50 different bins. By setting `density=True`, we're not plotting the *count* of darts, but the *proportion* in each bin. You can see that the shape of our stacked-up samples (the blue bars) almost perfectly recreates the original red "hill" ($f(x)$) we were aiming for. This confirms we successfully generated samples that follow the $\sin x$ curve.